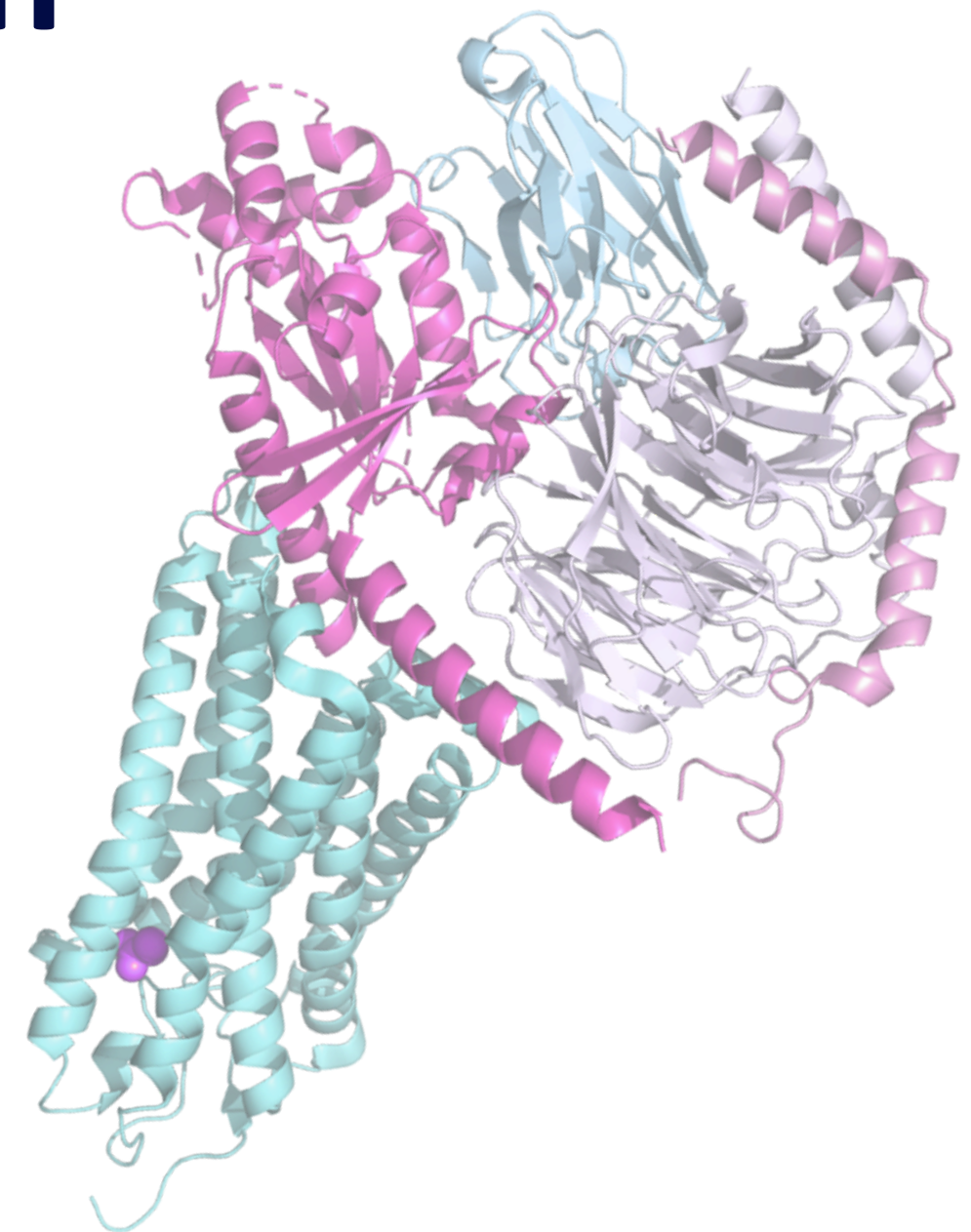


From molecules to perception

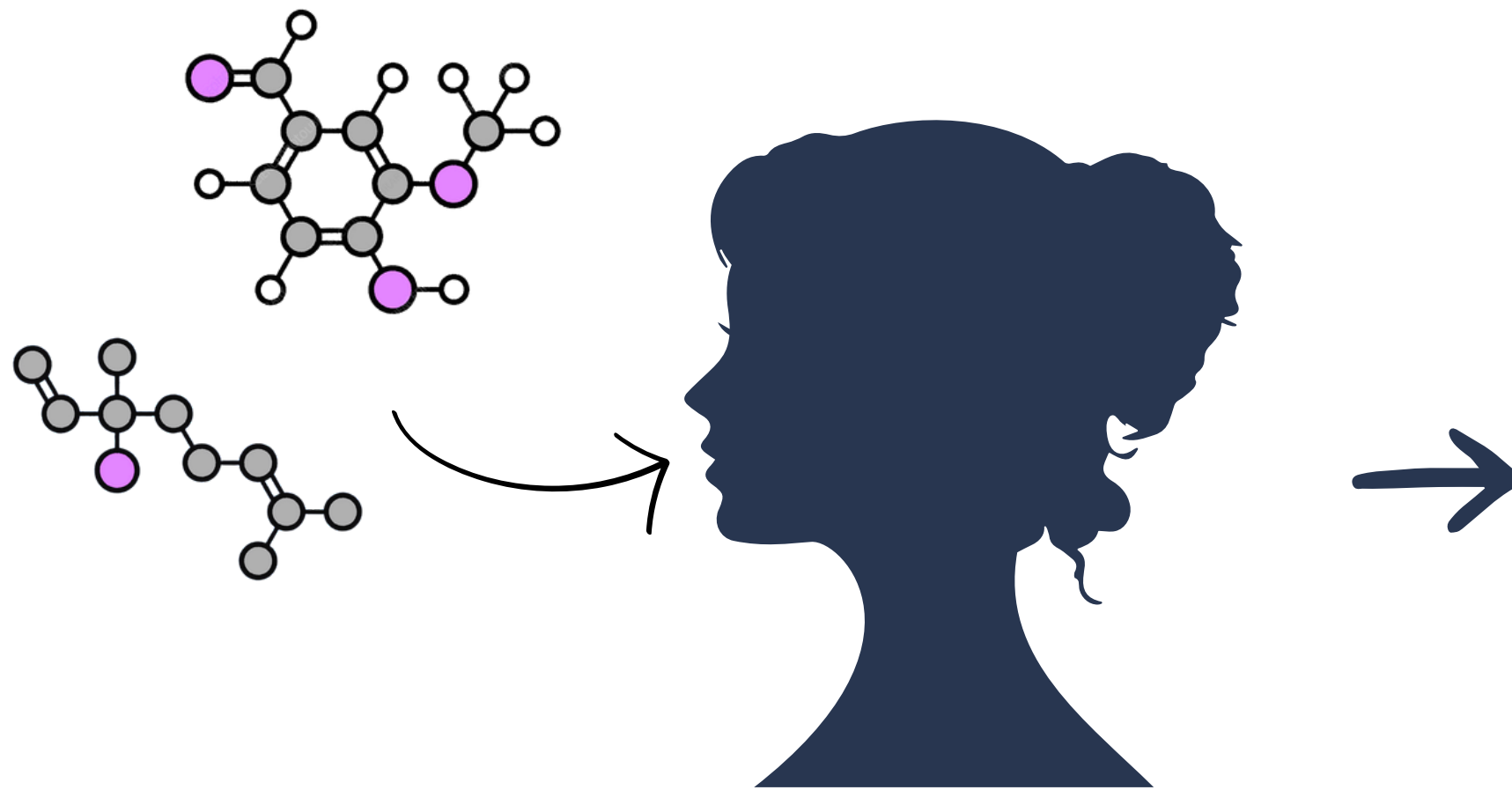
IMBB Comparative Genomics
Lab

Stella Tsiourva

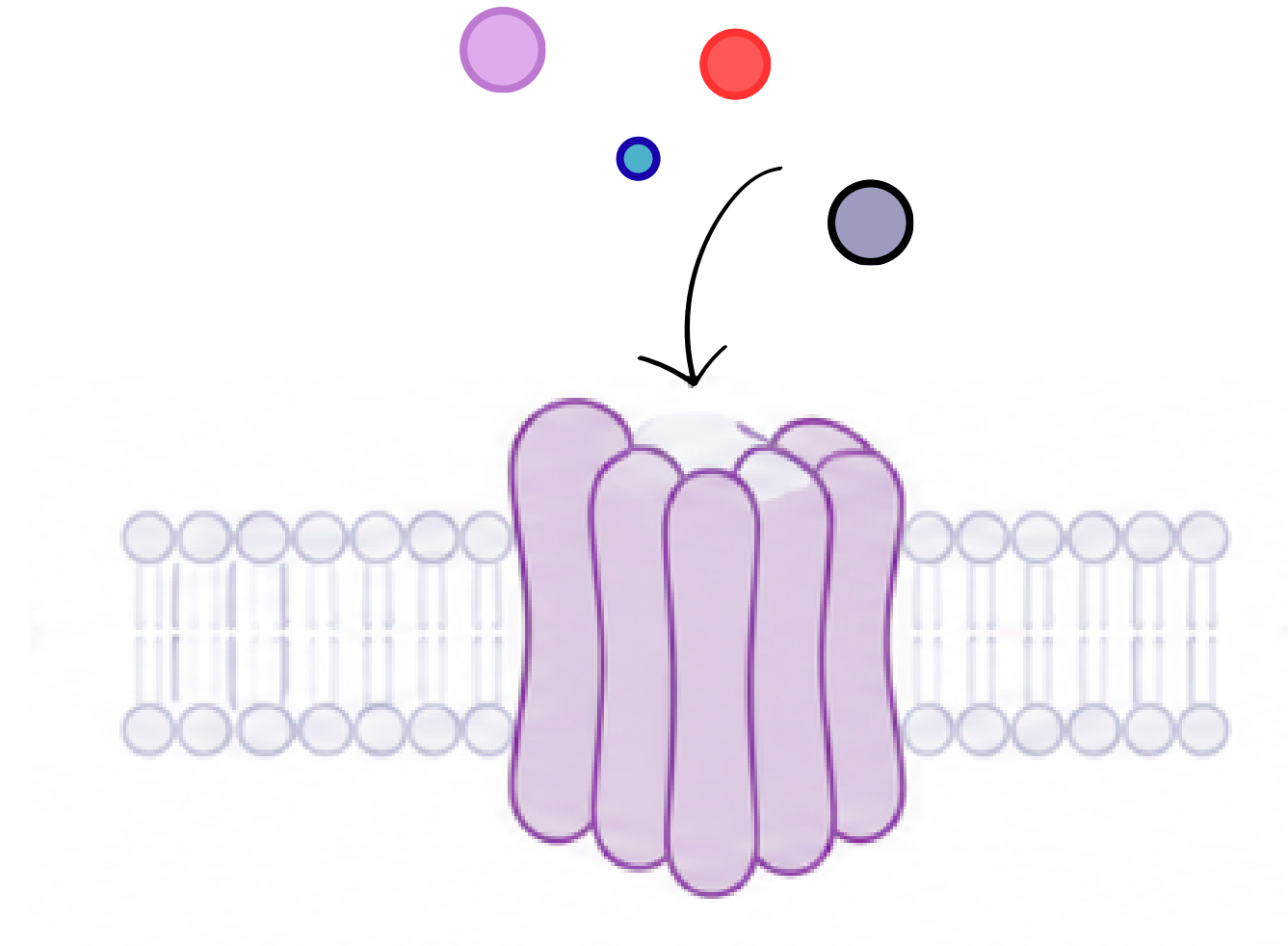
Heraklion, 2026



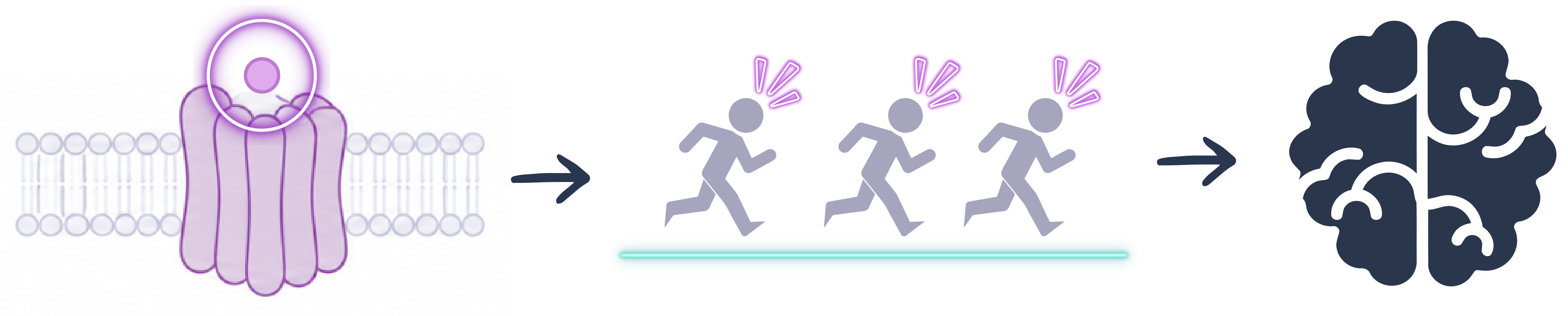
Smell begins as chemistry



Volatile odorants reach the
mucus



Binding to an olfactory
receptor (CPCR)

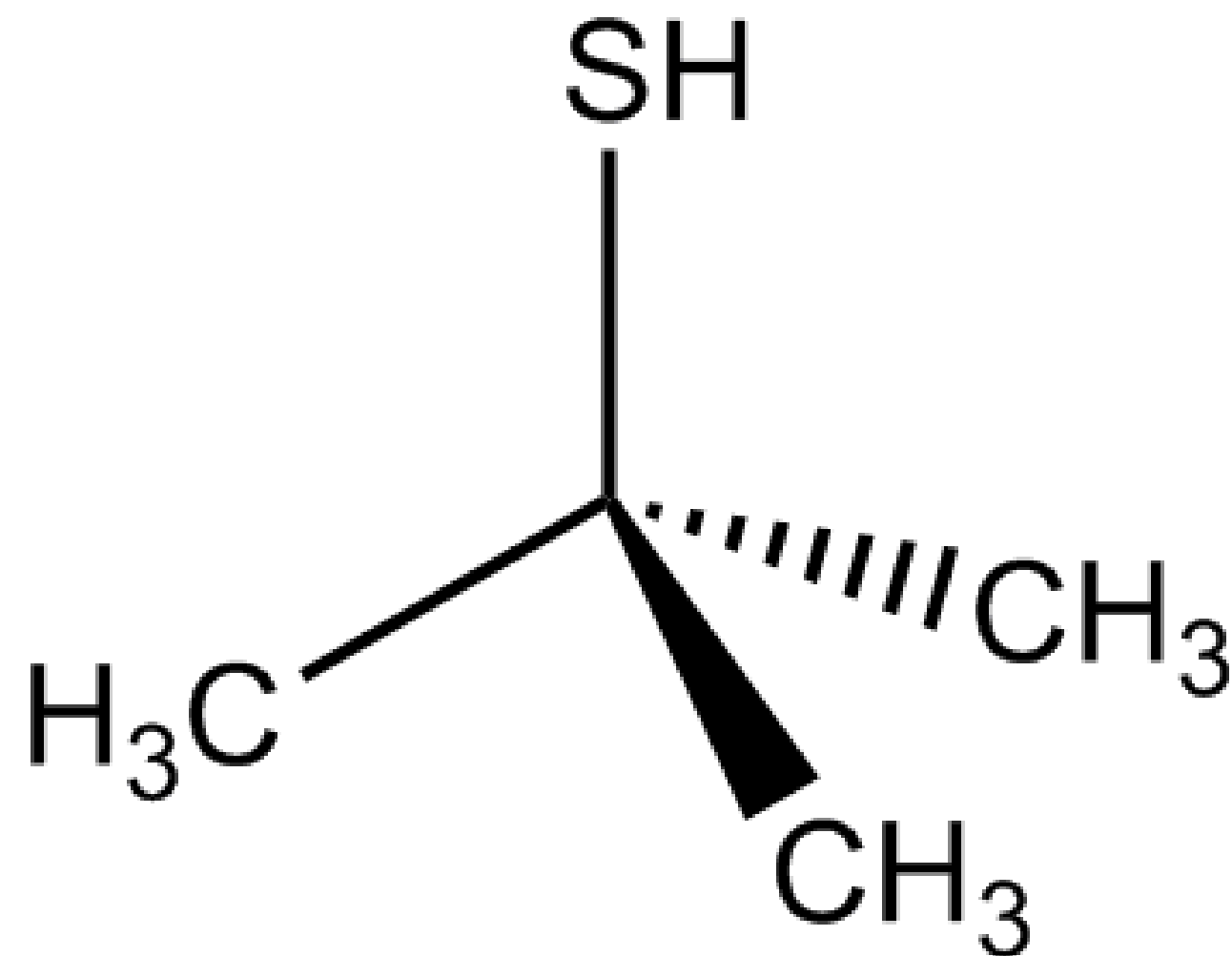


Neuron generates
electrical singnals

Signals are procced in the
brain and precieved as a
smell

What makes a molecule smell?

- * Volatility
- * Solubility
- * Chemical structure



MW= 90.19 Da

logP=2.22

The Data

M2OR

754

Molecules

575

With odor tags

139

Odor descriptors

RESEARCH QUESTION

What molecular changes drive discontinuities between chemical structure and odor perception?

Methodology

1.

Describe
molecules

2.

Compare

3.

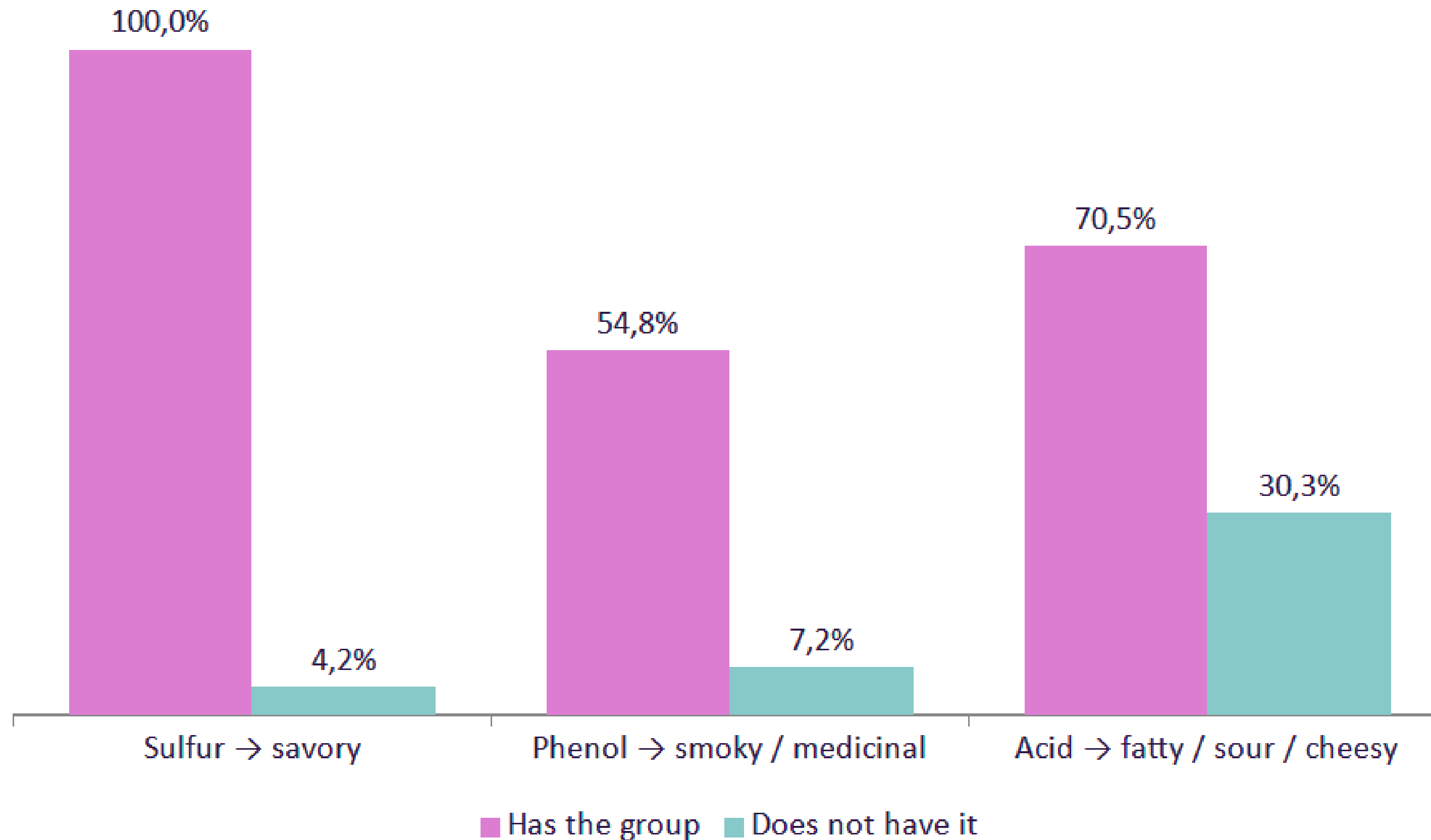
Find
discontinuities

4.

Explain

Morgan's fingerprints clustering

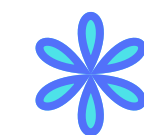
% of molecules carrying the target odor



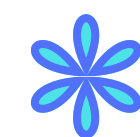
Odor similarity →



Structural similarity →

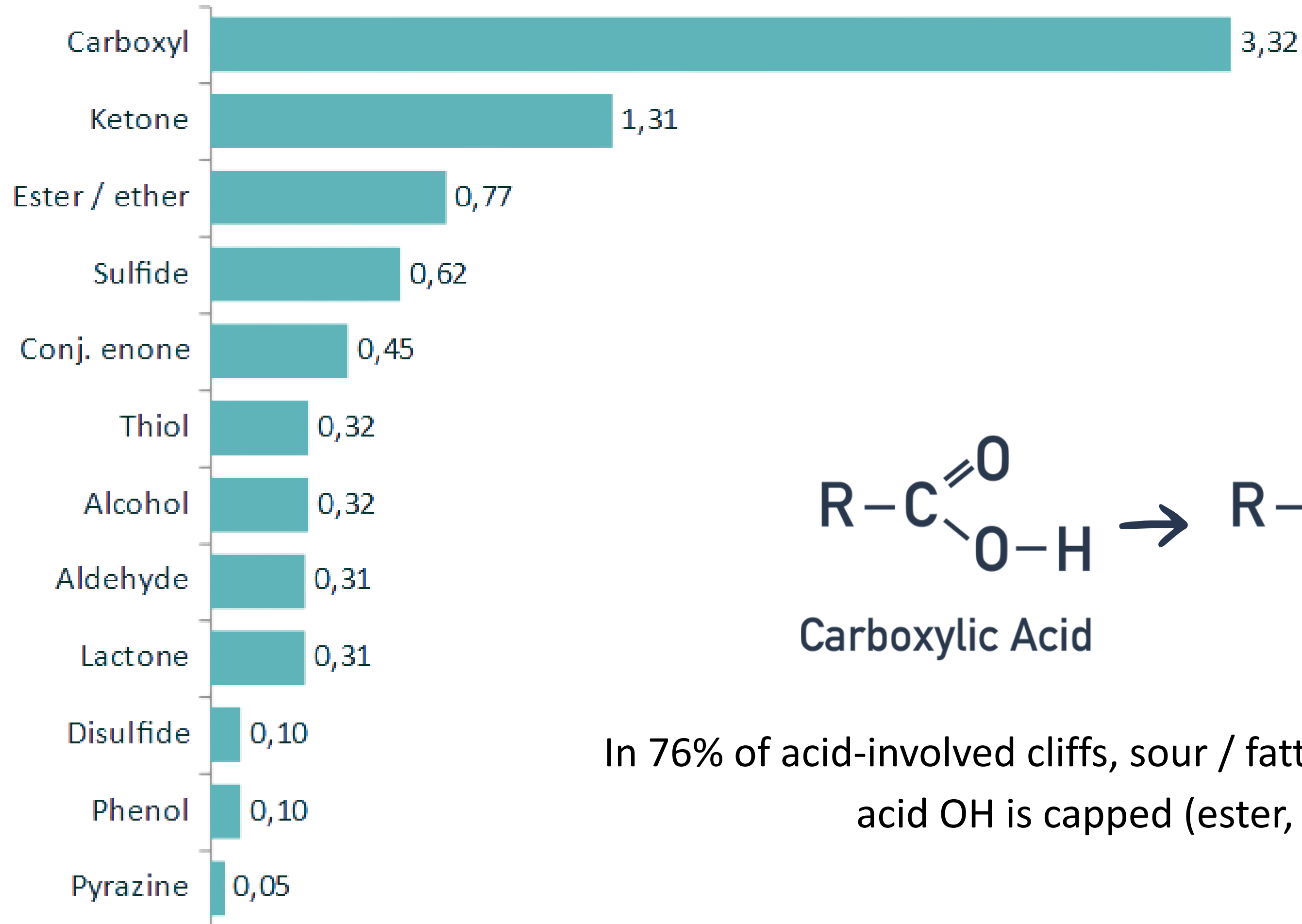


Comparison for all
165,025 pairs

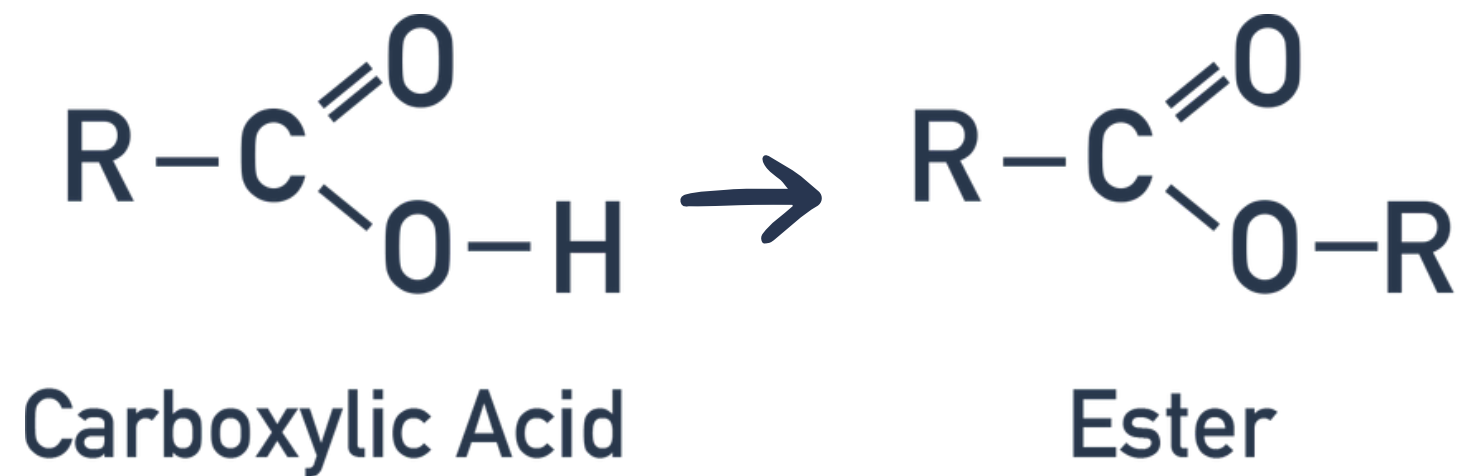


Tanimoto similarity ≥ 0.5
Jaccard odor similarity ≤ 0.1

Cliffs per molecule carrying the group



563 cliffs



In 76% of acid-involved cliffs, sour / fatty turns fruity once the acid OH is capped (ester, ketone).

Chain's length role

171 of 563 cliffs differ mainly in chain length or size — not in a functional group.

Chain length → a separate driver

Ethyl octanoate

CCCCCCCC(=O)OCC

apricot, musty, winey, pear, brandy



Ethyl nonanoate

CCCCCCCCC(=O)OCC

natural, tropical, fatty, oily, grape

Stereochemistry: invisible to fingerprints

22 pairs
of enantiomers

10 pairs
of E/Z isomers

All show odor divergence!



Same atoms, same bonds but a different smell.

Is it possible to predict the odor basis of the structure ?

Not right now...



The smallest changes can flip a smell

The mapping from molecular structure to odor perception is not simple or universally predictable.

Thank you!!!!

